# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The moment of inertia of a uniform circular disc of radius ' $R$ ' and mass ' $M$ ' about an axis passing from the edge of the disc and normal to the disc is

**(A)** $\mathrm{MR}^2$

**(B)** $\frac{1}{2}\mathrm{MR}^2$

**(C)** $\frac{3}{2}\mathrm{MR}^2$

**(D)** $\frac{7}{2}\mathrm{MR}^2$

**Q.2**

Moment of inertia does not depend upon

**(A)** distribution of mass

**(B)** axis of rotation

**(C)** point of application of force

**(D)** None of these

**Q.3**

Moment of inertia of a circular wire of mass $M$ and radius $R$ about its diameter is

**(A)** $\mathrm{MR}^2/2$

**(B)** $\mathrm{MR}^2$

**(C)** $2\mathrm{MR}^2$

**(D)** $\mathrm{MR}^2/4$

**Q.4**

Moment of inertia of a hollow cylinder of mass $\mathrm{M}$ and radius $\mathrm{r}$ about its own axis is

**(A)** $\frac{2}{3}\mathrm{Mr}^2$

**(B)** $\frac{2}{5}\mathrm{Mr}^2$

**(C)** $\frac{1}{3}\mathrm{Mr}^2$

**(D)** $\mathrm{Mr}^2$

**Q.5**

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Which of the following has the highest moment of inertia when each of them has the same mass and the same outer radius

**(A)** a ring about its axis, perpendicular to the plane of the ring

**(B)** a disc about its axis, perpendicular to the plane of the ring

**(C)** a solid sphere about one of its diameters

**(D)** a spherical shell about one of its diameters

**Q.6**

Radius of gyration of a body depends upon

**(A)** axis of rotation

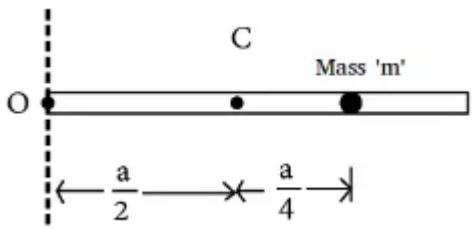
**(B)** translational motion

**(C)** shape of the body

**(D)** area of the body

**Q.7**

A particle of mass $m$ is attached to a thin uniform rod of length $\mathrm{a}$ and mass $4m$. The distance of the particle from the centre of mass of the rod is $a/4$. The moment of inertia of the combination about an axis passing through $\mathrm{O}$ normal to the rod is




**(A)** $\frac{64}{48}ma^2$

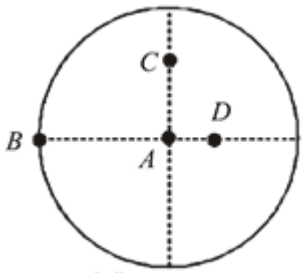
**(B)** $\frac{91}{48}ma^2$

**(C)** $\frac{27}{48}ma^2$

**(D)** $\frac{51}{48}ma^2$

**Q.8**

The moment of inertia of a uniform circular disc (figure) is maximum about an axis perpendicular to the disc and passing through




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**(A)** $B$

**(B)** $C$

**(C)** $D$

**(D)** $A$

**Q.9**

One solid sphere $A$ and another hollow sphere $B$ are of same mass and same outer radii, Their moments of inertia about their diameters are respectively $I_A$ and $I_B$, such that

**(A)** $\mathrm{I_A = I_B}$

**(B)** $\mathrm{I_A > I_B}$

**(C)** $\mathrm{I_A < I_B}$

**(D)** $\mathrm{I_A/I_B = d_A/d_B}$

**Q.10**

The moment of inertia of a uniform circular disc of radius ' $R$ ' and mass ' $M$ ' about an axis passing from the edge of the disc and normal to the disc is

**(A)** $\mathrm{MR^2}$

**(B)** $\frac{1}{2}\mathrm{MR^2}$

**(C)** $\frac{3}{2}\mathrm{MR^2}$

**(D)** $\frac{7}{2}\mathrm{MR^2}$

**Q.11**

Moment of inertia depends upon

**(A)** distribution of mass

**(B)** axis of rotation

**(C)** mass of body

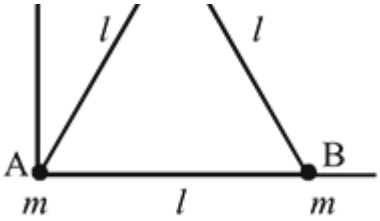
**(D)** All of these

**Q.12**

Three particles, each of mass $m$ gram, are situated at the vertices of an equilateral triangle $\mathrm{ABC}$ of side $\ell\mathrm{cm}$ (as shown in the figure). The moment of inertia of the system about a line AX perpendicular to $\mathrm{AB}$ and in the plane of $\mathrm{ABC}$, in gram-cm $^2$ units will be




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**(A)** $\frac{3}{2}\,\mathrm{m}\ell^2$

**(B)** $\frac{3}{4}\,\mathrm{m}\ell^2$

**(C)** $2\,\mathrm{m}\ell^2$

**(D)** $\frac{5}{4}\,\mathrm{m}\ell^2$

**Q.13**

The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axis in the plane of the ring is

**(A)** $1:\sqrt{2}$

**(B)** $1:3$

**(C)** $2:1$

**(D)** $\sqrt{5}:\sqrt{6}$

**Q.14**

The moment of inertia of a thin uniform rod of mass $\mathrm{M}$ and length $\mathrm{L}$ about an axis passing through its midpoint and perpendicular to its length is $\mathrm{I}_0$. Its moment of inertia about an axis passing through one of its ends and perpendicular to its length is

**(A)** $I_0 + ML^2/2$

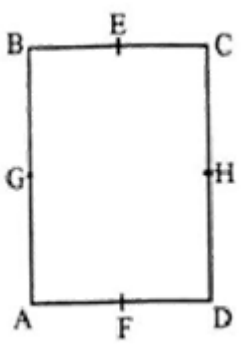
**(B)** $I_0 + ML^2/4$

**(C)** $I_0 + 2\mathrm{ML}^2$

**(D)** $\mathrm{I}_0 + \mathrm{ML}^2$

**Q.15**

The moment of inertia of the rectangular plate $\mathrm{ABCD}, (\mathrm{AB} = 2\mathrm{BC})$ is minimum along the axis




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**(A)** $\mathrm{GH}$

**(B)** $\mathrm{EF}$

**(C)** $\mathrm{BC}$

**(D)** $\mathrm{AC}$

**Q.16**

Four point masses, each of value $\mathrm{m}$, are placed at the corners of a square $\mathrm{ABCD}$ of side $\ell$. The moment of inertia of this system about an axis passing through $\mathrm{A}$ and parallel to $\mathrm{BD}$ is

**(A)** $2\ \mathrm{m}\ell^2$

**(B)** $\sqrt{3}\ \mathrm{m}\ell^2$

**(C)** $3\ \mathrm{m}\ell^2$

**(D)** $\mathrm{m}\ell^2$

**Q.17**

Consider a uniform square plate of side 'a' and mass ' $\mathrm{m}$ '. The moment of inertia of this plate about an axis perpendicular to its plane and passing through one of its corners is

**(A)** $\frac{5}{6}\mathrm{ma}^2$

**(B)** $\frac{1}{12}\mathrm{ma}^2$

**(C)** $\frac{7}{12}\mathrm{ma}^2$

**(D)** $\frac{2}{3}\mathrm{ma}^2$

**Q.18**

Point masses $1, 2, 3$ and $4\ \mathrm{kg}$ are lying at the points $(0,0,0), (2,0,0), (0,3,0)$ and $(-2,-2,0)$ respectively. The moment of inertia of this system about $X$ -axis will be

**(A)** $43\ \mathrm{kg\ m}^2$

**(B)** $34\ \mathrm{kg\ m}^2$

**(C)** $27\ \mathrm{kg\ m}^2$

**(D)** $72\ \mathrm{kg\ m}^2$

**Q.19**

From a solid sphere of mass $\mathrm{M}$ and radius $\mathrm{R}$ a cube of maximum possible volume is cut. Moment of inertia of cube about an axis passing through its center and perpendicular to one of its faces is :

**(A)** $\frac{4\mathrm{MR}^2}{9\sqrt{3}\pi}$

**(B)** $\frac{4\mathrm{MR}^2}{3\sqrt{3}\pi}$

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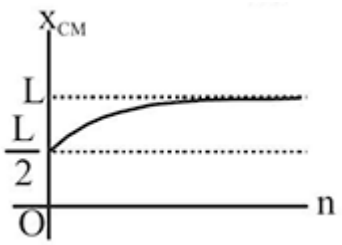
**(C)** $\frac{MR^2}{32\sqrt{2}\pi}$

**(D)** $\frac{MR^2}{16\sqrt{2}\pi}$

**Q.20**

A thin rod of length ' $L$ ' is lying along the $x$ -axis with its ends at $x=0$ and $x=L$. Its linear density (mass/ length) varies with $x$ as $k\left(\frac{x}{L}\right)^n$, where $n$ can be zero or any positive number. If the position $x_{CM}$ of the centre of mass of the rod is plotted against ' $n$ ', which of the following graphs best approximates the dependence of $x_{CM}$ on $n$ ?

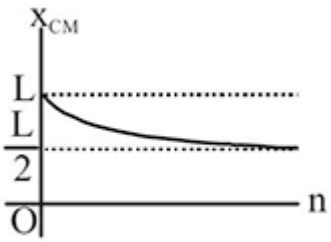
**(A)**



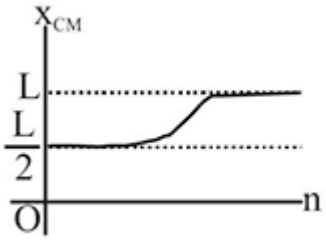

**(B)**




**(C)**




**(D)**




## Answers & Solutions

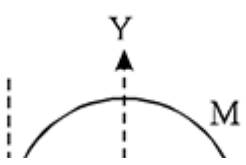
**Q.1 Answer:**

$\frac{3}{2}MR^2$

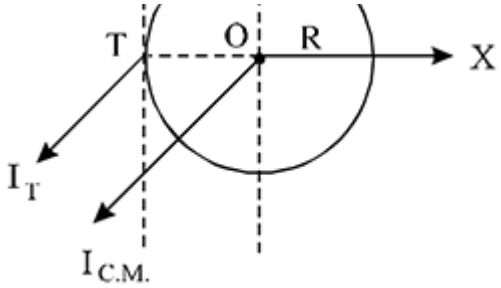
**Solution:**

**M.I. of a uniform circular disc of radius 'R' and mass 'M' about an axis passing through C.M. and normal to the disc is**

$I_{C.M}=\frac{1}{2}MR^2$




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**From parallel axis theorem** $I_T = I_{C.M.} + MR^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$

**Q.2 Answer:**

point of application of force

**Solution:**

**No solution available**

**Q.3 Answer:**

$MR^2/2$

**Solution:**

**Perp axis theorem** $I_x + I_y = I_z^2$

$\Rightarrow \quad 2I = MR^2 \quad$ **So** $\quad I = \frac{MR^2}{2}$

**Q.4 Answer:**

$Mr^2$

**Solution:**

**Moment of inertia of a hollow cylinder of mass** $M$ **and radius** $r$ **about its own axis is** $Mr^2$

**Q.5 Answer:**

a ring about its axis, perpendicular to the plane of the ring

**Solution:**

**The correct option is (1)**

**A ring about its axis, perpendicular to the plane of the ring.**

**(1) Moment of inertia of a ring about its axis and perpendicular to its plane** $= Mr^2$

**(2) Moment of inertia of a disc about its axis and perpendicular to its plane** $= \frac{1}{2}Mr^2 = 0.5Mr$

**(3) Moment of inertia of a solid sphere about one of its diameter** $= \frac{2}{5}Mr^2 = 0.4Mr^2$

**(4) Moment of inertia of a spherical shell about one of its diameter** $= \frac{2}{3}Mr^2 = 0.66Mr^2$

**Therefore, the moment of inertia of the ring is highest.**

**Q.6 Answer:**

axis of rotation

**Solution:**

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**Radius of gyration of a body depends on the axis of rotation.**

**Q.7 Answer:**

$\frac{91}{48}ma^2$

**Solution:**

**Moment of inertia**

$= m\left(\frac{3a}{4}\right)^2 + m_1\frac{a^2}{3}$

**For the centre of rod**

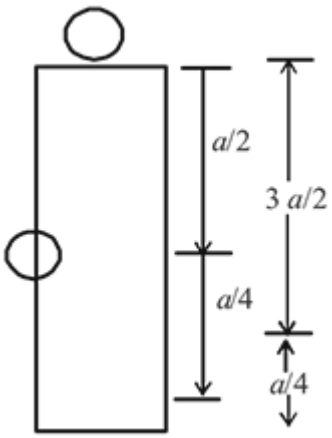
$\left(\frac{m_1a^2}{12} + \frac{m_1a^2}{4}\right) = \frac{m_1a^2}{3}$

$\therefore \quad m_1 = 4m$

**Total** $I = m\left(\frac{3a}{4}\right)^2 + \frac{4ma^2}{3}$

$= \frac{9ma^2}{16} + \frac{4ma^2}{3}$

$= \frac{(27+64)}{48}ma^2 = \frac{91}{48}ma^2$




**Q.8 Answer:**

$B$

**Solution:**

**According to parallel axis theorem of the moment of Inertia**

$$I = I_{\text{cm}} + md^2$$

$d$ **is maximum for point** $B$ **so** $I_{\text{max}}$ **about** $B$**.**

**Q.9 Answer:**

$\mathrm{I_A} < \mathrm{I_B}$

**Solution:**

**Given; mass and radius of both the spheres are same. Now as moment of inertia of solid sphere,** $\mathrm{I_A} = \frac{2}{5}\mathrm{MR}^2$

**Moment of inertia of hollow sphere,** $\mathrm{I_B} = \frac{2}{3}\mathrm{MR}^2$

$\implies \mathrm{I_A} < \mathrm{I_B}$

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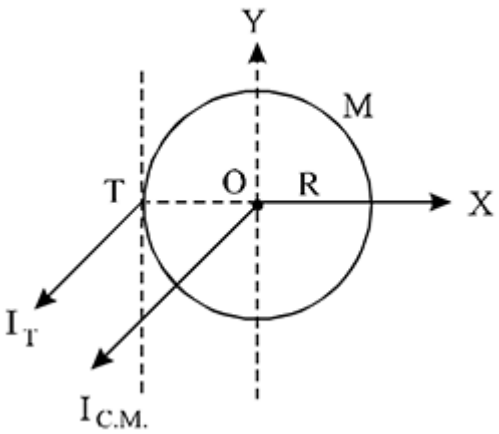
**Q.10 Answer:**

$\frac{3}{2}MR^2$

**Solution:**

**M.I. of a uniform circular disc of radius 'R' and mass 'M' about an axis passing through C.M. and normal to the disc is**

$I_{C.M} = \frac{1}{2}MR^2$




**From parallel axis theorem** $I_T = I_{C.M.} + MR^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$

**Q.11 Answer:**

All of these

**Solution:**

**Moment of inertia is also known as the rotational inertia. When masses are in linear motion, its rotational analog is known as rotational inertia. The moment of inertia provides the relationship for dynamics of rotational motion. The moment of inertia can be calculated with respect to the axis of rotation of the particles.**

**Thus, moment of inertia in other terms is known as the mass distribution of the particle with respect to the axis of rotation. The distribution of the particle from the axis of rotation is also dependent on the shape and size of the object. Thus, the moment of inertia of the object depends on the mass, axis of rotation and shape and size of the body.**

**Q.12 Answer:**

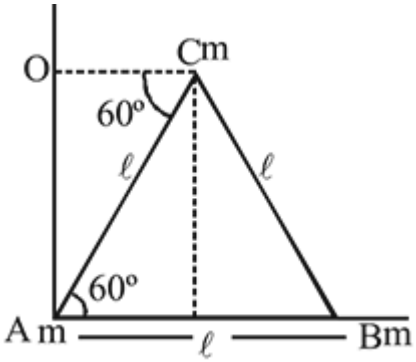
$\frac{5}{4}\ m\ell^2$

**Solution:**

$$\begin{aligned} I_{AX} &= m(AB)^2 + m(OC)^2 \\ &= m\ell^2 + m\left(\ell\cos 60^\circ\right)^2 \\ &= m\ell^2 + m\ell^2/4 = 5/4\ m\ell^2 \end{aligned}$$



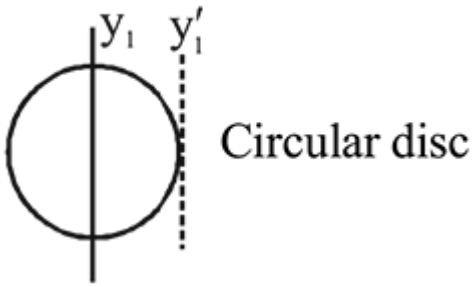
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**Q.13 Answer:**

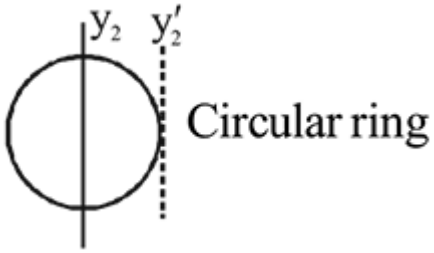
$\sqrt{5} : \sqrt{6}$

**Solution:**




$\mathrm{I_{y_1}} = \frac{\mathrm{MR}^2}{4}$

$\therefore \mathrm{I'_{y_1}} = \frac{\mathrm{MR}^2}{4} + \mathrm{MR}^2 = \frac{5}{4}\mathrm{MR}^2$




$I_{y_2} = \frac{MR^2}{2}$

$\therefore I'_{y_2} = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$

$I'_{y_1} = MK_1^2, I'_{y_2} = MK_2^2$

$\therefore \frac{K_1^2}{K_2^2} = \frac{I'_{y_1}}{I'_{y_2}} \Rightarrow K_1 : K_2 = \sqrt{5} : \sqrt{6}$

**Q.14 Answer:**

$I_0 + ML^2/4$

**Solution:**

**By theorem of parallel axes,** $\mathrm{I} = \mathrm{I_{cm}} + \mathrm{Md}^2$

$\mathrm{I} = \mathrm{I_0} + \mathrm{M(L/2)}^2 = \mathrm{I_0} + \mathrm{ML}^2/4$

**Q.15 Answer:**

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EF

**Solution:**

**The distribution of mass about axis EF is minimum so radius of gyration is minimum and therefore moment of inertia is minimum about EF**

**Q.16 Answer:**

$3\ \mathrm{m}\ell^2$

**Solution:**

$\mathrm{I_{nn'}}$ **= M.I due to the point mass at** $\mathrm{B+}$

**M.I due to the point mass at** $\mathrm{D}$ **+ M.I due to the point mass at** $C. = 2 \times \mathrm{m}\left(\frac{\ell}{\sqrt{2}}\right)^2 + \mathrm{m}(\sqrt{2}\ell)^2 = \mathrm{m}\ell^2 + 2\ \mathrm{m}\ell^2 = 3\ \mathrm{m}\ell^2$

**Q.17 Answer:**

$\frac{2}{3}\mathrm{ma}^2$

**Solution:**

$I_{\mathrm{nn'}} = \frac{1}{12}M\left(a^2 + a^2\right) = \frac{Ma^2}{6}$

**Also,** $\mathrm{DO} = \frac{\mathrm{DB}}{2} = \frac{\sqrt{2}\mathrm{a}}{2} = \frac{\mathrm{a}}{\sqrt{2}}$

**According to parallel axis theorem** $\mathrm{I_{mm'}} = \mathrm{I_{nn'}} + \mathrm{M}\left(\frac{\mathrm{a}}{\sqrt{2}}\right)^2 = \frac{\mathrm{Ma}^2}{6} + \frac{\mathrm{Ma}^2}{2}$

$= \frac{\mathrm{Ma}^2 + 3\mathrm{Ma}^2}{6} = \frac{2}{3}\mathrm{Ma}^2$

**Q.18 Answer:**

$43\ \mathrm{kg\ m}^2$

**Solution:**

**Moment of inertia of the whole system about the axis of rotation will be equal to the sum of the moments of inertia of all the particles.**

$= 0 + 0 + 27 + 16 = 43\ \mathrm{kg\ m}^2$

**Q.19 Answer:**

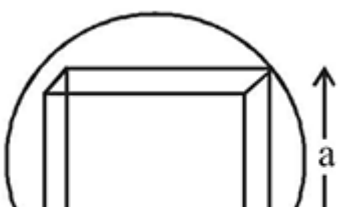
$\frac{4\mathrm{MR}^2}{9\sqrt{3}\pi}$

**Solution:**

**Here a**

$= \frac{\frac{4}{3}\pi\mathrm{R}^3}{\left(\frac{2}{\sqrt{3}}\mathrm{R}\right)^3} = \frac{\sqrt{3}}{2}\pi \cdot \quad \mathrm{M'} = \frac{2\mathrm{M}}{\sqrt{3}\pi}$

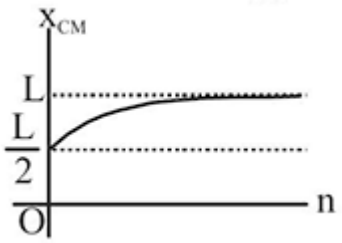
**Moment of inertia of the cube about the given axis,** $I = \frac{M'a^2}{6} = \frac{\frac{2M}{\sqrt{3}\pi} \times \left(\frac{2}{\sqrt{3}}R\right)^2}{6} = \frac{4MR^2}{9\sqrt{3}\pi}$




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**Q.20 Answer:**




**Solution:**

**When $n = 0, x = k$ where $k$ is a constant. This means that the linear mass density is constant. In this case the centre of mass will be at the middle of the rod i.e., at $L/2$. Therefore (C) is ruled out**

**- $n$ is positive and as its value increases, the rate of increase of linear mass density with increase in $x$ increases. This shows that the centre of mass will shift towards that end of the rod where $n = L$ as the value of $n$ increases. Therefore graph (B) is ruled out.**

**- The linear mass density $\lambda = k\left(\frac{x}{L}\right)^n$**

**Here $\frac{x}{L} \leq 1$**

**With increase in the value of $n$, the centre of mass shift towards the end $x = L$ such that first the shifting is at a higher rate with increase in the value of $n$ and then the rate decreases with the value of $n$.**

**These characteristics are represented by graph (A).**

$$x_{CM} = \frac{\int_0^L x dm}{\int_0^L dm} = \frac{\int_0^L x(\lambda dx)}{\int_0^L \lambda dx} = \frac{\int_0^L k\left(\frac{x}{L}\right)^n x dx}{\int_0^L k\left(\frac{x}{L}\right)^n dx}$$

$$= \frac{k\left[\frac{x^{n+2}}{(n+2)L^n}\right]_0^L}{\left[\frac{kx^{n+1}}{(n+1)L^n}\right]_0^L} = \frac{L(n+1)}{n+2}$$

**For $n = 0, x_{CM} = \frac{L}{2}; n = 1,$**

$x_{CM} = \frac{2L}{3}; n = 2, x_{CM} = \frac{3L}{4}; \ldots$

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